Discrete Eigenvalues: Bound States

Why Trapped Quantum Systems Have Allowed Energies

Quantum Chemistry Notes

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July 9, 2026

Big Idea

A bound quantum system does not allow any random energy. The allowed energies come from solving the Schrödinger equation with boundary conditions and normalisation.

1 Where are we now?

In the previous note, I looked at uncertainty, wavefunctions, operators, and the Schrödinger equation.

The main equation we reached was

$$\hat{H}\psi = E\psi.$$

This is the time-independent Schrödinger equation.

Now the question is:

Question

Why do bound quantum systems have only certain allowed energies?

This is one of the most important ideas in quantum mechanics.

In early quantum theory, physicists had to guess quantisation rules. But in wave mechanics, quantisation comes naturally from the allowed wavefunctions.

2 Why do we care about eigenvalues?

In quantum mechanics, physical quantities are represented by operators.

When an operator acts on a special state and gives back the same state multiplied by a number, that state is called an eigenstate.

For example,

$$\hat{A}\psi = a\psi.$$

Here, ψ is the eigenstate and a is the eigenvalue.

The operator we care about now is the Hamiltonian $\hat{H}$, which represents the total energy of the system.

So we write

$$\hat{H}\psi = E\psi.$$

This means that ψ is a state with definite energy E.

Note to Self

The equation $\hat{H}\psi = E\psi$ alone does not automatically mean the energy is discrete. Free particles can also have energy eigenstates, but their energies can be continuous. Discrete energy appears when the particle is bound and the wavefunction has to satisfy strict conditions.

3 The time-independent Schrödinger equation

The full Schrödinger equation is time-dependent:

$$i\hbar\frac{\partial\Psi(x,t)}{\partial t} = \hat{H}\Psi(x,t).$$

If the potential does not depend on time, we can separate the wavefunction into a space part and a time part:

$$\Psi(x,t) = \psi(x)\tau(t).$$

The time part has the solution

$$\tau(t) = Ae^{-iEt/\hbar}.$$

So the full wavefunction can be written as

$$\Psi(x,t) = \psi(x)e^{-iEt/\hbar}.$$

The space part satisfies

$$\hat{H}\psi = E\psi.$$

For one dimension,

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + V(x)\psi = E\psi.$$

This is the equation we will use for bound states.

Big Idea

The time-independent Schrödinger equation finds the allowed spatial wavefunctions and their corresponding energies.

4 What is a bound state?

When we say something is bound, we mean that the object is trapped or localised.

For example:

- a particle trapped inside a potential well,
- an electron bound to an atom,
- atoms vibrating inside a molecule,
- particles trapped near a stable equilibrium point.

In quantum mechanics, a bound state means that the wavefunction is normalisable.

In one dimension,

$$\int_{-\infty}^{\infty} |\psi(x)|^2 \, dx < \infty.$$

After normalisation, we write

$$\int_{-\infty}^{\infty} |\psi(x)|^2 \, dx = 1.$$

This means that the total probability of finding the particle somewhere is 1.

For a bound state, the wavefunction cannot just spread infinitely like a free particle wave. It must become small far away from the region where the particle is trapped.

So, roughly,

$$\psi(x) \to 0 \quad \text{as} \quad |x| \to \infty.$$

Big Idea

A bound state is a normalisable quantum state. The particle is not free to run away.

5 Why do bound states give discrete energy?

To find the wavefunction of a bound state, we need to satisfy a few conditions.

1. We must solve the time-independent Schrödinger equation.
2. The wavefunction must satisfy the boundary conditions.
3. The wavefunction must be finite.
4. The wavefunction must be single-valued.
5. The wavefunction must usually be continuous.
6. Its derivative must also usually be continuous, unless the potential has an infinite jump.
7. The wavefunction must be normalisable.

Because of these conditions, not every wavefunction is allowed.

Most possible mathematical solutions get rejected.

A good intuition is a guitar string. The string cannot vibrate in any random way. It can only form certain standing waves. Those standing waves have certain allowed frequencies.

Similarly, a bound quantum particle can only have certain allowed wavefunctions.

Since energy is connected to the wavefunction through

$$\hat{H}\psi = E\psi,$$

only certain energies are allowed.

$$\text{Boundary conditions + normalisation} \implies \text{allowed wavefunctions} \implies \text{allowed energies.}$$

6 A finite potential well

Let us now look at the rough shape of a bound-state wavefunction.

Consider a one-dimensional finite potential well:

$$V(x) = \begin{cases} 0, & 0 < x < a, \\ V_0, & x < 0 \text{ or } x > a. \end{cases}$$

For a bound state in this well,

$$0 < E < V_0.$$

Inside the well, where $0 < x < a$, the particle is in the classically allowed region.

The Schrödinger equation becomes

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} = E\psi.$$

The solution is sinusoidal:

$$\psi(x) = A\sin(kx) + B\cos(kx),$$

where

$$k = \frac{\sqrt{2mE}}{\hbar}.$$

Outside the well, the energy of the particle is less than the potential energy:

$$E < V_0.$$

This is the classically forbidden region.

There, the wavefunction becomes exponential. For $x > a$, the acceptable solution is

$$\psi(x) = Ce^{-\kappa(x-a)},$$

where

$$\kappa = \frac{\sqrt{2m(V_0 - E)}}{\hbar}.$$

We choose the decaying exponential because the wavefunction must not blow up as $x \to \infty$.

For $x < 0$, the acceptable solution is

$$\psi(x) = De^{\kappa x}.$$

So the rough form is

$$\psi(x) = \begin{cases} De^{\kappa x}, & x < 0, \\ A\sin(kx) + B\cos(kx), & 0 < x < a, \\ Ce^{-\kappa(x-a)}, & x > a. \end{cases}$$

Big Idea

Inside the well, the wavefunction oscillates. Outside the well, it decays. This is the basic shape of a bound-state wavefunction in a finite well.

The full solution comes from matching ψ and $\frac{d\psi}{dx}$ at the boundaries $x = 0$ and $x = a$.

Those matching conditions only work for certain energies.

That is why the finite well has discrete bound-state energies.

Note to Self

For the finite well, the energy values usually do not come from a simple formula. They come from equations that must be solved graphically or numerically. The infinite well gives the cleaner formula.

7 A note on notation: why some books use $u(x)$

Some books write the spatial wavefunction as $u(x)$ instead of $\psi(x)$.

In one dimension, this is usually just notation:

$$u(x) = \psi(x).$$

So if a book writes

$$u(x) = A\sin(kx) + B\cos(kx),$$

it is just writing the spatial wavefunction with a different symbol.

But in the three-dimensional spherical case, $u(r)$ has a more specific meaning. There, one often writes

$$u(r) = rR(r),$$

where $R(r)$ is the radial part of the wavefunction.

We will come back to this briefly later.

8 Infinite square well as the clean limit

The finite well gives the realistic picture.

The infinite square well gives the cleanest formula.

For an infinite square well,

$$V(x) = \begin{cases} 0, & 0 < x < a, \\ \infty, & x < 0 \text{ or } x > a. \end{cases}$$

Since the walls are infinite, the particle cannot exist outside the well.

So,

$$\psi(0) = 0, \qquad \psi(a) = 0.$$

Inside the well, the solution is

$$\psi_n(x) = A \sin\left(\frac{n\pi x}{a}\right),$$

where

$$n = 1, 2, 3, \ldots$$

After normalisation,

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right).$$

The allowed energies are

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2}.$$

This formula clearly shows discreteness.

The energy cannot be any random value. It must depend on the integer n.

Big Idea

The energy is discrete because only waves with an integer number of half-wavelengths fit inside the box.

9 Zero-point energy

One very important thing is that the lowest energy is not zero.

For the infinite square well, the lowest energy is

$$E_1 = \frac{\pi^2\hbar^2}{2ma^2}.$$

This is an example of zero-point energy.

A confined quantum particle cannot be perfectly still.

If it were perfectly still, its momentum would be exactly zero. But since the particle is confined, its position is restricted. This creates uncertainty in position, and therefore uncertainty in momentum.

Roughly,

$$\Delta x \sim a.$$

By the uncertainty principle,

$$\Delta x \Delta p \geq \frac{\hbar}{2}.$$

So,

$$\Delta p \gtrsim \frac{\hbar}{2a}.$$

This means the particle must have some kinetic energy.

Connection

This connects directly to the previous note on uncertainty. Confinement creates momentum uncertainty, and momentum uncertainty creates non-zero kinetic energy.

10 Probability density for bound states

The wavefunction itself is not directly the probability.

The probability density is

$$|\psi(x)|^2.$$

If the particle is in the state $\psi(x)$, then the probability of finding it between x and $x + dx$ is

$$|\psi(x)|^2 dx.$$

For the infinite square well,

$$|\psi_n(x)|^2 = \frac{2}{a}\sin^2\left(\frac{n\pi x}{a}\right).$$

Some important points:

1. The wavefunction $\psi(x)$ can be positive or negative.
2. The probability density $|\psi(x)|^2$ is always non-negative.
3. Points where $\psi(x) = 0$ are called nodes.
4. Higher energy states have more nodes.
5. For finite wells, there is a small probability of finding the particle outside the well.

For a finite well, the probability outside the well is

$$P_{\text{outside}} = \int_{-\infty}^{0} |\psi(x)|^2\, dx + \int_{a}^{\infty} |\psi(x)|^2\, dx.$$

This probability is small, but it is not zero.

This is why tunnelling is possible.

The particle does not have enough classical energy to cross the barrier, but the wavefunction still exists in the classically forbidden region.

Note to Self

We are not going deeply into tunnelling here. But the finite well already shows the seed of the idea.

11 Why we now look at the linear harmonic oscillator

Before going further, it is useful to look at the linear harmonic oscillator.

At first, it may feel like this is just another example. But it is much more important than that.

The harmonic oscillator is one of the most useful bound-state models in physics.

The potential is

$$V(x) = \frac{1}{2} m\omega^2 x^2.$$

This potential traps the particle near $x = 0$.

The time-independent Schrödinger equation becomes

$$-\frac{\hbar^2}{2m}\frac{d^2\psi}{dx^2} + \frac{1}{2} m\omega^2 x^2 \psi = E\psi.$$

The allowed energies are

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega,$$

where

$$n = 0, 1, 2, 3, \ldots$$

So the energy levels are discrete and equally spaced.

The ground state energy is

$$E_0 = \frac{1}{2}\hbar\omega.$$

Again, the lowest energy is not zero.

The ground state wavefunction is

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-m\omega x^2/2\hbar}.$$

This is a Gaussian. It is largest near $x = 0$ and decays as x moves away from the centre.

The higher wavefunctions have the general form

$$\psi_n(x) = N_n H_n \left(\sqrt{\frac{m\omega}{\hbar}} x \right) e^{-m\omega x^2/2\hbar}.$$

Here, H_n are Hermite polynomials and N_n is a normalisation constant.

We do not need to focus on Hermite polynomials right now.

The main idea is:

$$\text{harmonic oscillator} \quad \Longrightarrow \quad \text{discrete energy + zero-point energy + normalisable wavefunctions.}$$

12 Why the harmonic oscillator appears everywhere

The harmonic oscillator is important because many potentials look like a parabola near a stable equilibrium point.

Suppose a potential has a minimum at $x = x_0$.

Near that minimum, we can approximate it using a Taylor expansion:

$$V(x) \approx V(x_0) + \frac{1}{2} V''(x_0)(x - x_0)^2.$$

This has the same form as the harmonic oscillator potential.

So whenever a particle is trapped near a stable minimum, the first approximation is often a harmonic oscillator.

This is why the harmonic oscillator appears in:

- vibrating molecules,
- atoms in solids,
- phonons,
- small oscillations around equilibrium,
- quantum field theory.

Big Idea

The harmonic oscillator is not just one example. It is the universal model for small bound motion near a stable equilibrium.

13 The three-dimensional case

In three dimensions, the wavefunction depends on three coordinates:

$$\psi = \psi(x, y, z).$$

The time-independent Schrödinger equation becomes

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(x, y, z)\psi = E\psi.$$

The normalisation condition becomes

$$\int |\psi(x, y, z)|^2 \, d^3r = 1.$$

The meaning is the same as in one dimension.

A bound state is still a normalisable state. The particle is still localised. The energy becomes discrete because only certain wavefunctions satisfy the Schrödinger equation, the boundary conditions, and normalisation.

The only difference is that the geometry becomes richer. Instead of just left and right, the particle can move in all directions.

14 Spherically symmetric potentials

A very important special case is when the potential depends only on the distance from the origin:

$$V(x, y, z) = V(r),$$

where

$$r = \sqrt{x^2 + y^2 + z^2}.$$

This is called a spherically symmetric potential.

This case is important because atoms are often treated this way.

For example, the hydrogen atom has the Coulomb potential

$$V(r) = -\frac{e^2}{4\pi\epsilon_0 r}.$$

When the potential is spherically symmetric, we use spherical coordinates:

$$(r, \theta, \phi).$$

The wavefunction can be separated into a radial part and an angular part:

$$\psi(r, \theta, \phi) = R(r) Y_\ell^m(\theta, \phi).$$

Here, $R(r)$ tells us how the wavefunction changes with distance from the centre.

The function $Y_\ell^m(\theta, \phi)$ describes the angular shape.

This is where quantum numbers like

$$n, \ell, m$$

come from.

Roughly:

$$n \quad \text{controls the energy and radial structure,}$$

$$\ell \quad \text{controls the angular momentum shape,}$$

$$m \quad \text{controls the orientation.}$$

This is also where atomic orbitals come from:

$$\ell = 0 \Rightarrow s\text{-orbitals},$$

$$\ell = 1 \Rightarrow p\text{-orbitals},$$

$$\ell = 2 \Rightarrow d\text{-orbitals}.$$

Sometimes, instead of using $R(r)$ directly, books define

$$u(r) = rR(r).$$

This turns the radial Schrödinger equation into a simpler one-dimensional-looking equation:

$$-\frac{\hbar^2}{2m}\frac{d^2u}{dr^2} + \left[V(r) + \frac{\hbar^2\ell(\ell+1)}{2mr^2}\right]u = Eu.$$

The extra term

$$\frac{\hbar^2\ell(\ell+1)}{2mr^2}$$

acts like an angular momentum barrier.

Note to Self

We are not deriving the spherical case fully now. For now, I just want to see how one-dimensional bound-state ideas lead naturally toward atoms and orbitals.

Connection

This is the bridge to quantum chemistry. Bound states in spherically symmetric potentials lead to atomic orbitals.

15 Quick Summary

Idea	Meaning	Why it matters
Eigenvalue equation	$\hat{H}\psi = E\psi$	Finds states with definite energy
Bound state	Normalisable, localised state	Particle is trapped instead of free
Boundary conditions	Conditions on ψ at walls or infinity	Reject most possible solutions
Discrete energy	Only certain E values are allowed	Comes from allowed wavefunctions
Finite well	Wavefunction leaks outside the well	Leads toward tunnelling
Infinite well	Clean model with exact energy formula	Shows standing-wave quantisation
Harmonic oscillator	Particle bound by parabolic potential	Universal model for small oscillations
Spherical potentials	Potentials depending only on r	Leads toward atoms and orbitals

Table 1: The main ideas behind bound states and discrete eigenvalues.

16 Final Takeaway

A bound state is a quantum state where the particle is trapped or localised.

Mathematically, this means the wavefunction must be normalisable.

For one dimension,

$$\int |\psi(x)|^2 \, dx = 1.$$

For three dimensions,

$$\int |\psi(x, y, z)|^2 \, d^3r = 1.$$

Bound states give discrete energies because only certain wavefunctions satisfy all the required conditions:

$$\text{Schrödinger equation} + \text{boundary conditions} + \text{normalisation}.$$

The finite well shows that the wavefunction can leak outside the classically allowed region.

The infinite well gives the clean formula

$$E_n = \frac{n^2\pi^2\hbar^2}{2ma^2}.$$

The harmonic oscillator gives

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega.$$

Both show that bound quantum systems do not allow arbitrary energies.

Big Idea

Discrete energy is not randomly assumed. It comes from the wavefunctions that are allowed to exist.

References and Reading Notes

- Standard quantum mechanics texts on bound states and the time-independent Schrödinger equation.
- Standard discussions of the finite square well, infinite square well, and linear harmonic oscillator.
- Notes on spherical symmetry and atomic orbitals from introductory quantum chemistry and quantum mechanics texts.